

EMILY Y. XU

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EDUCATION

California Institute of Technology (Caltech)

Expected Graduation: 12/2026

B.S., Electrical Engineering - GPA: 4.2/4.0

Pasadena, CA

Coursework: Deterministic Analysis of Systems and Circuits (EE44), Mathematics of Electrical Engineering (EE55), Physics of Electrical Engineering (EE40), Electronic Systems Lab (EE45), Electronic Systems (EE13), Data Structures/Algorithms (CS2), Software Design (CS3), Machine Learning/Data Mining (CS156, CS155), Deep Learning (EE150), Digital Logic and Embedded Systems (EE10), Calculus I, II, III, Linear Algebra, Differential Equations, Probability and Statistics, Physics E/M, Ethical AI (PI45)

EXPERIENCE

Summer Undergraduate Research Fellow (Machine Learning)

Incoming

Autonomous Robotics and Control Laboratory, Advised by Prof. Soon-jo Chung and Dr. Thomas Berrueta

Pasadena, CA

- Implemented Domain-Adversarially Invariant Meta-Learning (DAIML) on adaptive throttle control for the Caltech Indy Autonomous Challenge vehicle. Developed vehicle simulations in C++ and ROS.

Summer Undergraduate Research Fellow (Physics/Software)

Jun 2024 – Aug 2024

Caltech Observational Cosmology, Advised by Prof. Sunil Golwala and Brandon Sandoval

Pasadena, CA

- Designed aluminum quasi-particle trapping capacitor fins for Kinetic Inductance Detector (KID) devices. Optimized design and analysis using circuit theory, algorithms, and quality factor research to improve phonon collection efficiency through current modeling in Sonnet. Presented to Fermilab and NASA JPL researchers.

Student Researcher (Physics), Canadian Team Captain

Feb 2022 – Aug 2022

International Young Physicists' Tournament (IYPT)

Toronto, ON | Timisoara, Romania

- Led a team of 5 to compete in a thesis-defense-style competition for 17 open-ended physics problems.
- Designed a three-axis movement system with Arduino for experimental automation. Developed MATLAB scripts for real-time control and electric potential data acquisition, integrating with Excel for data storage.
- Performed electromagnetic simulations to investigate the Double Layer and Debye length in COMSOL for theoretical verification and conducted data analysis and visualization using Python. Achieved 95% modeling accuracy.

PROJECTS

Photodiode Array for Real-Time Hand Gesture Recognition ☞ | *MATLAB Neural Net Pattern Recognition*

- Designed and built a custom 2D array of photodiodes, trans-impedance amplifiers, and an Arduino-based analog acquisition pipeline.
- Developed a MATLAB-based machine learning classifier to detect gestures from the photodiode intensity map.

Infinite Vertical Platformer Video Game ☞ | *C, SDL, POSIX API*

- Developed a full stack single and multiplayer video game inspired by Doodle Jump from scratch in C, using SDL for rendering.
- Implemented custom physics engine, sprites, keyboard controls, an intelligent enemy AI, UI, and unit testing.
- Utilized POSIX API for system-level programming, optimizing memory management and algorithm efficiency.
- Created multiple intermediate game prototypes (Pacman, N-Body simulation, Crossy Road) to validate physics engine and graphics.

LLM Stock Market Analysis Tool ☞ | *Python, Beautiful Soup, PostgreSQL, Docker, Flask, Plotly*

- Built a dynamic web application using Flask and HTML for real-time stock data visualization and AI-powered market insights
- Implemented web scraping with Beautiful Soup, data storage with PostgreSQL, and containerization within Docker, while designing interactive Plotly graphs based on user-input stock tickers and date ranges.
- Integrated a fine-tuned LLM using Gemini API for market analysis, optimizing with prompt engineering and context-aware caching; increased efficiency by 60% and reduced costs compared to GPT API

LEADERSHIP

Mission Lead, Caltech Air and Outer Space

Oct 2023 – Nov 2024

National Aeronautics and Space Association (NASA), Caltech Air and Outer Space (CAOS)

Pasadena, CA

- Directed a team of 10 Caltech students in crafting the mission concept and background research review for a 26-page proposal.
- Secured \$150,000 in funding to advance research on the proposed inflation technology, performed fluids simulations in COMSOL.

TECHNICAL SKILLS

Programming Languages: Python, Java, C/C++, SQL, ABEL (HDL), L^AT_EX

Frameworks & Libraries: TensorFlow, Scikit-learn, PyTorch, Flask, Dash, Plotly, Matplotlib, OpenCV, Keras

Tools: LTSpice, MATLAB, PostgreSQL, Docker, Arduino, SolidWorks, COMSOL, Sonnet, DipTrace, Git, VS Code

Skills: Circuit Design, Machine Learning, Deep Learning, Data Analysis, Data Mining, Scientific Writing, Research Methodologies, OOP

AWARDS

International Young Physicists' Tournament: Bronze 2022

Canadian Young Physicists' Tournament: 2x Gold 2022, 2023

Carnegie Mellon University Women's Mathematics Competition: Overall Third Place 2022